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CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Technology (B.Tech)

Semester – II University Examination May 2010

B.Tech. (CL/EC/ME)

B.Tech. (CE/EE/IT)

PY - 101 Engineering Physics

Date: 17.05.10, Monday **Time: 12:30 pm to 03:30 pm** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION - I

Q. 1 (a) Calculate the reverberation time for a room 20 m x 15 m and 8 m high. The room contains 200 upholstered seats and half of them are occupied. The absorption coefficients of the wall, ceiling, floor, the empty seat and occupied seat are 0.09, 0.04, 0.06, 0.43 and 0.64 respectively. 3

(b) What is laser? How does laser differ from ordinary light? 4

Q. 2 (a) The force acting on an object of mass "m" traveling at velocity "v" in a circle of radius "r" is given by $F = mv^2/r$. The measurements are recorded as m = (3.5 ± 0.1) kg, v = (20 ± 1) m/s and r = (12.5 ± 0.5) m. Find the maximum possible (i) fractional error and (ii) percentage error in the measurement of force. 4

(b) Explain the design considerations involved for achieving rooms of good acoustical quality. 5

(c) Define Einstein's coefficients and obtain the relations between them. 5

OR

Q. 2 (a) "Any error in the measurement can be minimized by using care and proper experimental techniques, but it cannot be eliminated." Justify the statement. 4

(b) Explain the acoustic grating method of determination of velocity of ultrasonic wave. 5

(c) Define pumping and population inversion in any laser action. The ratio of population of two energy levels at 300 K is 10^{-30}. Find the wavelength of the radiation emits. Given that: h = 6.63 x 10^{-34} J s and k_B = 1.38 x 10^{-23} J/K. 5

Q.3 (a) Discuss the properties of ultrasound and on the basis of that explain various applications of ultrasound in different fields. 7

(b) Describe the construction and working of Nd-YAG laser with suitable energy level diagram. Give any two applications of this laser. 7

OR

Q.3 (a) Define magnetostriction and piezoelectricity. Explain with a neat circuit, the generation of ultrasonic waves using piezoelectric oscillator. 7

(b) Define numerical aperture, acceptance angle and relative refractive index of an optical fiber. The refractive index of the core is 1.5 and the fractional refractive index between the core and cladding is 1.8%. Estimate (i) numerical aperture, (ii) acceptance angle, (iii) critical angle at the core cladding interface and (iv) velocity of light in the core and cladding. 7

SECTION - II

Q.4 (a) Define lattice, basis and unit cell. 3

(b) Discuss the important property changes that occur in materials when they change from normal to superconducting state. 4

Q.5 (a) Explain lattice parameters of a unit cell? Lead has a FCC structure. Give details about coordination number, number of atoms per unit cell, lattice constant and atomic packing factor. 7

(b) What are nanostructured materials? Discuss any five size dependant physical properties of nanomaterials with examples. 7

OR

Q.5 (a) Define x-ray crystallography. Name the seven crystal systems and give the geometrical characteristics of each crystal systems. 7

(b) What are the salient features of nanotechnology? Write any five applications of nanomaterials in the field of engineering and industry. 7

Q.6 (a) Derive Braggs Law. An X-ray beam of wavelength (λ_1) undergoes a first order Bragg reflection at a Bragg angle of 23^0. X-rays of wavelength 97 pm undergo third order reflection at a Bragg angle of 60^0. Assuming that the two beams are reflected from the same set of planes, calculate interplanar spacing d and λ_1. 7

(b) Give an account of phenomena superconductivity. Discuss the classification of super conductors based on their behavior in an applied magnetic field and the type of coolants used. 7

OR

Q.6 (a) What are the significances of Miller indices? Draw the following planes in a cubic lattice: (1 1 0), (1 1 2) and (1 0 1). 7

(b) Explain how the Hall voltage in Hall effect experiment helps one to determine the type, density and mobility of charge carriers. The Hall voltage and conductivity for the metal sodium is 0.001 mV and 2.09 $\Omega^{-1}m^{-1}$, measured at I = 100 mA, B = 2 weber/m^2 and the width of the specimen = 0.05mm. Calculate the number of carriers per m^3 and mobility of the electrons in sodium. 7